



Original Article

Ergothioneine oxidation in the protection against high-glucose induced endothelial senescence: Involvement of SIRT1 and SIRT6



Nunzia D'Onofrio^a, Luigi Servillo^a, Alfonso Giovane^a, Rosario Casale^a, Milena Vitiello^a, Raffaele Marfella^b, Giuseppe Paolisso^b, Maria Luisa Balestrieri^{a,*}

^a Department of Biochemistry, Biophysics and General Pathology, Second University of Naples, Naples, Italy

^b Department of Medical, Surgical, Neurological, Metabolic and Aging Sciences, Second University of Naples, Naples, Italy

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ABSTRACT

Ergothioneine (Egt), the betaine of 2-mercapto-L-histidine, is a dietary antioxidant protecting against many diseases, including cardiovascular disease (CVD), through a redox mechanism different from alkylthiols. Here, experiments were designed to evaluate the mechanisms underlying the beneficial effect of Egt against hyperglycaemia-induced senescence in endothelial cells. To this end, cells were incubated with increasing concentrations of Egt (0.01–1.00 mM) for 12 h followed by incubation for 48 h with high-glucose (25 mM). Cell evaluation indicated that viability was not affected by mM concentrations of Egt and that the high-glucose cytotoxicity was prevented with the highest efficacy at 0.5 mM Egt. The cytoprotective effect of Egt was paralleled by reduced ROS production, cell senescence, and, interestingly, the formation of hercynine (EH), a betaine we recently found to be produced during the Egt oxidation pathway. Notably, the Egt beneficial effect was exerted through the upregulation of sirtuin 1 (SIRT1) and sirtuin 6 (SIRT6) expression and the downregulation of p66Shc and NF-κB. SIRT1 activity inhibition and SIRT6 gene silencing by small interfering RNA abolished the protective effect of Egt against the high-glucose-induced endothelial senescence. These data provide the first evidence of the Egt ability to interfere with endothelial senescence linked to hyperglycaemia through the regulation of SIRT1 and SIRT6 signaling, thus further strengthening the already assessed role of these two histone deacetylases in type 2 diabetes.

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1. Introduction

Endothelial dysfunction is a multifaceted disorder associated with a number of pathophysiological processes strictly linked to risk factors of cardiovascular disease (CVD), such as hypertension, atherosclerosis, or diabetes [1,2]. Oxidative stress, the common

denominator underlying endothelial dysfunction in CVD, results in a high accumulation of hydrogen peroxide (H₂O₂) and superoxide (O₂^{•−}), and a reduced nitric oxide (NO) bioavailability, thus driving endothelial cells (EC) towards senescence and apoptotic processes which interfere with the mechanisms of endothelial repair and regeneration [1,3]. In particular, the enhanced generation of reactive oxygen species (ROS) and the inflammatory activation characterize the functional alteration of the endothelium during hyperglycaemia, a key feature of diabetes and its vascular complications [4–6].

The hyperglycaemia-induced endothelial dysfunction is mediated by the downregulation of sirtuin 1 (SIRT1) and sirtuin 6 (SIRT6) [4,7–9]. SIRT1 protects blood vessels from hyperglycaemia-induced endothelial dysfunction through the downregulation of p66Shc expression [10]. In EC, SIRT1 prevents hydrogen peroxide-induced premature senescence by deacetylating the tumor suppressor p53 and protects from hyperglycaemia-induced endothelial dysfunction through the transcriptional regulation of endothelial nitric oxide synthase (eNOS) and the downregulation of p66Shc expression, a unique protein isoform of SHC (Src homology 2 domain containing) transforming protein 1 [11]. In EC,

Abbreviations: CVD, cardiovascular disease; EC, endothelial cells; Egt, ergothioneine; EH, hercynine; eNOS, endothelial nitric oxide synthase; ESH, 2-mercapto-histidine betaine; ESO₂H, ergothioneine sulfinic acid; ESOH, ergothioneine sulfenic acid; ESSE, ergothioneine disulfide; GSH, glutathione; hGluc, high-glucose; ICAM-1, intercellular adhesion molecule-1; IL-1β, interleukin-1β; MCP-1, monocyte chemoattractant protein-1; NADH, nicotinamide adenine dinucleotide; NF-κB, nuclear factor-kappaB; NO, nitric oxide; OCTN1, organic cationic transporter 1; PKCβII, protein kinase c βII; ROS, reactive oxygen species; SHC, Src homology 2 domain containing; SIRT1, sirtuin 1; SIRT6, sirtuin 6; TNF-α, tumor necrosis factor-α; VCAM-1, vascular cell adhesion molecule-1; EIC, extracted ion chromatogram; ICC, ion charge control; ESI, electrospray ionization; FIA, flow injection analysis; HPLC, high-performance liquid chromatography; MS/MS, tandem mass; MRM, multiple reaction monitoring

* Corresponding author.

E-mail address: marialuisa.balestrieri@unina2.it (M.L. Balestrieri).

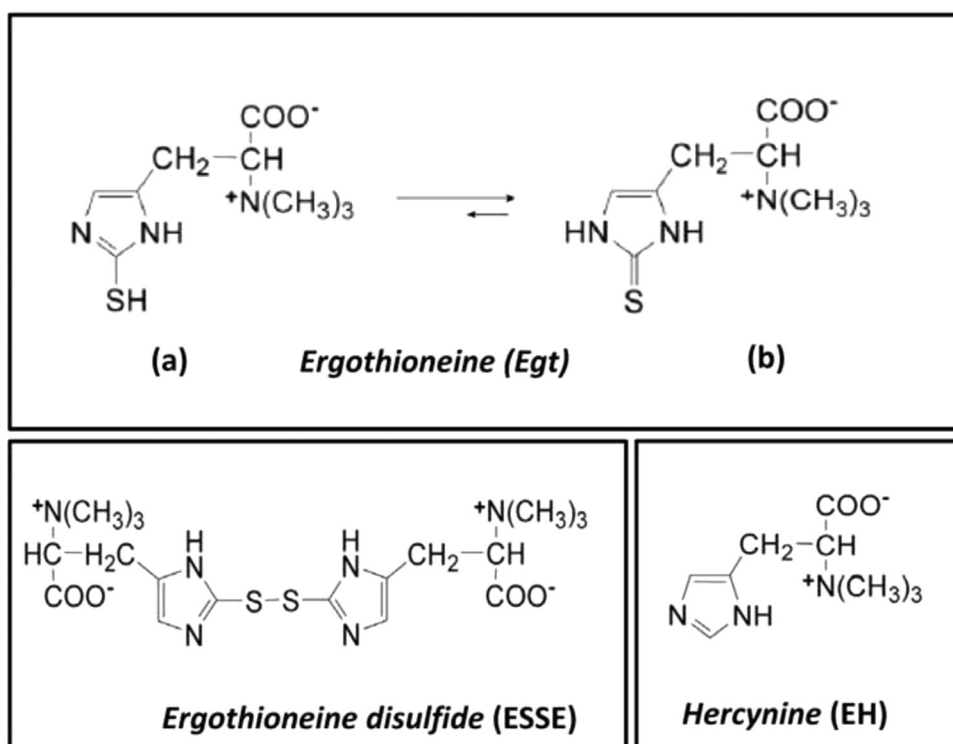


Fig. 1. Chemical structure of ergothioneine (Egt), its disulfide (ESSE), and hercynine (EH). Tautomeric equilibrium between thiol (a) and thione (b) form of ergothioneine.

the atheroprotective effect of SIRT1 is also exerted through the activation of eNOS and the inhibition of NF- κ B activity [11]. Several SIRT1 modulators, including resveratrol, resveratrol derivative, and stachydrine showed a beneficial effect on EC dysfunction, through positive regulation of SIRT1 pathway and reduction of superoxide production [12–15]. Indeed, to date, the pharmacological modulation of SIRT1 for the treatment of endothelial dysfunction and vascular-aging disorders is under intensive pre-clinical and clinical research [11].

Ergothioneine (Egt), 2-mercaptohistidine betaine (ESH), is an unusual compound containing the 2-thioimidazole moiety, widely distributed both in the plant and animal kingdoms (Fig. 1). Plants absorb Egt via symbiotic associations between their roots and soil fungi, whereas mammals acquire Egt solely through the diet. Indeed, this naturally occurring water soluble compound, firstly isolated from the ergot fungus (*Claviceps purpurea*) [16], is synthesized only by Actinomycetales and non-yeast like fungi [17–19].

At physiological pH, Egt is mainly in the thione form (Fig. 1). Although many reports describe the Egt chemical properties and synthesis [19], the Egt biosynthetic pathway from *Mycobacterium smegmatis* has been only recently fully reconstituted *in vitro* showing the involvement of Egt sulfenic acid (ESOH) as an intermediate [20]. This biosynthetic profile is consistent with the behavior, identified by Servillo and co-workers [21], of Egt under oxidative conditions, in which ESOH acts as a key intermediate. The Egt disulfide (ESSE) (Fig. 1), formed by Egt in the presence of oxidants, is unstable at physiological pH and undergoes a progressive decomposition. Specifically, ESSE is first hydrolyzed into Egt (ESH) and the highly reactive ESOH that disproportionates producing one molecule of ESH and one of the Egt sulfenic acid (ESO₂H). Successively, the highly unstable ESO₂H irreversibly hydrolyzes into sulfurous acid and hercynine (EH), the betaine of histidine [21].

The fact that Egt can be partially reformed from its disulfide ESSE, without the need of reducing agents, and then oxidized again in a new redox cycle, explains the absolute difference from

alkylthiols, such as cysteine and GSH [21].

The endothelial uptake of Egt occurs through the organic cation transporter novel type-1 (OCTN-1) and silencing this specific cell membrane carrier with small interfering RNA abolished the antioxidant and cytoprotective effects of Egt [22]. Owing to the specific carrier, Egt introduced by diet mainly accumulates up to millimolar range also in cells and tissues predisposed to inflammation and oxidative stress [19,23]. Notably, Egt itself is not toxic to EC up to 10 mM [22,24]. A reduced expression of adhesion molecule VCAM-1, ICAM-1 and E-selectin has been observed following treatment of EC with Egt at concentrations of 0.1–0.3 mM [24]. A higher concentration of Egt (1 mM) was effective in reducing monocyte binding to EC [24] and in suppressing reactive oxygen species (ROS) production and cell death in EC exposed to pyrogallol, xanthine oxidase plus xanthine, and high-glucose [22].

The protective action of Egt against CVD during ischemia/reperfusion is achieved through the reduction of ferrylmyoglobin, the modulation of proinflammatory cytokines, and iron chelation [25–27]. Controversial are the findings about the role of Egt in diabetes, since diabetic patients have been reported to have elevated levels of Egt which affects insulin and glucagon storage through chelation of zinc [28,29]. On the other hand, no significant elevation of Egt has been observed in rat model of alloxan-induced diabetes [30]. Moreover, Egt chronic treatment of rats with streptozocin-induced diabetes ameliorated the response to acetylcholine in arteries [22] and Egt supplementation of diabetic pregnant rats reduced diabetic embryopathy, probably through the modulation of hyperglycemia-mediated oxidative stress [31].

While Egt effects on ROS-induced cytotoxicity and vascular responsiveness have been elucidated both *in vitro* and *in vivo* models, our understanding of the molecular mechanism(s) underlying these actions is still relatively limited. Moreover, the recent findings on the peculiar redox behavior of Egt, which widen the scenario of its antioxidant cellular properties, prompted us to investigate the molecular mechanism(s) through which Egt acts in the prevention of CVD, specifically, in the protection of EC against

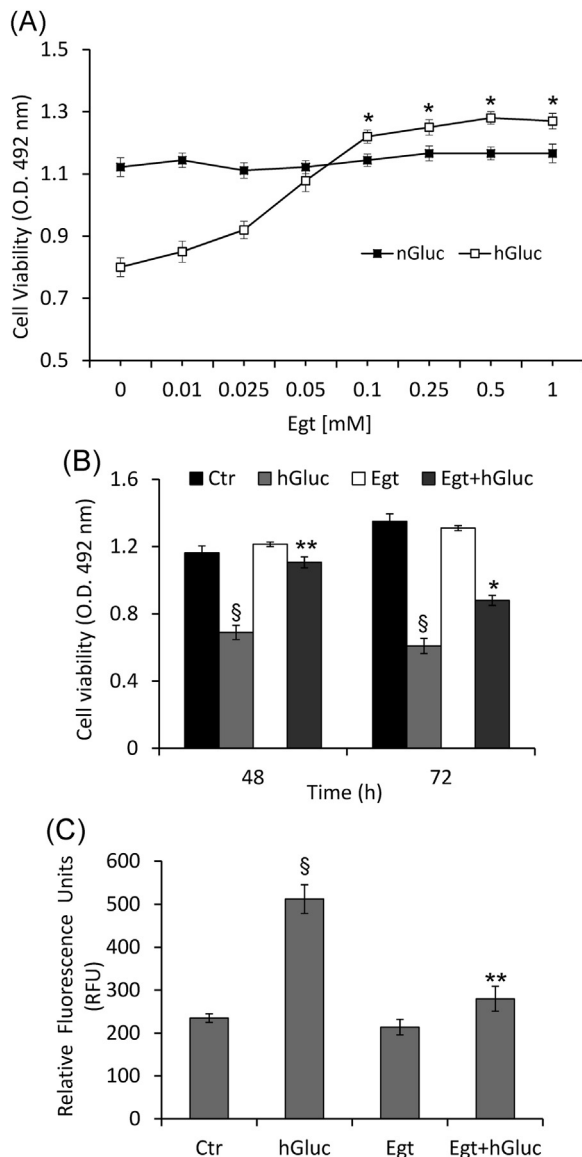


Fig. 2. Antioxidant and cytoprotective effect of ergothioneine. (A) EC viability was determined by XTT assay on cells pre-treated for 12 h with different Egt concentrations, followed by treatment for 48 h with high-glucose (25 mM), hGluc, or control media containing normal-glucose (nGluc); * $P < 0.05$ vs. nGluc, § $P < 0.01$ vs. nGluc. (B) Cell viability was determined after 48 h and 72 h of treatment with Egt (0.5 mM), high-glucose (hGluc), high-glucose in the presence of Egt (0.5 mM) (Egt+hGluc), or control media (Ctrl) containing nGluc; * $P < 0.05$ vs. hGluc, ** $P < 0.01$ vs. hGluc, and § $P < 0.01$ vs. Ctrl. (C) Intracellular ROS production is reported as the amount of DCF fluorescence intensity after 48 h of treatment with Egt (0.5 mM), high-glucose (hGluc), Egt plus high-glucose (Egt+hGluc), or control media (Ctrl); ** $P < 0.01$ vs. hGluc, and § $P < 0.01$ vs. Ctrl. (A, B, C) Values are the mean $\pm$ SD of four independent experiments.

the high-glucose damages. To this end, after having evaluated the Egt concentrations not toxic to EC, we investigated its effects on high-glucose induced cytotoxicity and senescence, deepening the possible relationship with SIRT1 and/or SIRT6 signaling.

2. Material and methods

2.1. Chemicals and antibodies

L-ergothioneine was purchased from Santa Cruz Biotechnology (Santa Cruz, CA, USA). Minimum Essential Medium (MEM, GIBCO, 11095-080), foetal bovine serum (FBS), and Pen-Strep were from

Gibco, Life Technologies). NEAA (Non-Essential Amino Acid solution 100 $\times$) was purchased from Lonza. XTT (sodium 30-[1-(phenylaminocarbonyl)-3,4-tetrazolium]-bis (4-methoxy-6-nitro) benzene sulfonic acid hydrate) assay was obtained from Roche. 2',7'-dichlorofluorescein diacetate (DCFH-DA) was purchased from Sigma-Aldrich Co. (St. Louis, MO, USA). β -galactosidase (SA- β -Gal) activity staining kit was obtained from Cell Signalling Technology (Danvers, MA, USA). Sirtinol was purchased from Sigma-Aldrich Co. (St. Louis, MO, USA). SIRT6 siRNA oligos set, control non targeting siRNA (NT-siRNA), and RNAiFectin™ transfection reagent were from Applied Biological Materials Inc. (Richmond, BC, Canada). Antibodies against SIRT1 (rabbit monoclonal, ab32441) and SIRT6 (rabbit polyclonal, ab62738) were obtained from Abcam (Cambridge, UK). Antibody against NF- κ B (rabbit monoclonal, C22B4) was purchased from Cell Signaling Technology (Danvers, MA). Antibody against gamma-tubulin (mouse monoclonal, GTU-88) was obtained from Sigma (St. Louis, MO, USA). Secondary antibodies Alexa Fluor® 488 and Alexa Fluor® 633 were obtained from Life Technologies Italia (Monza, Italy). All other solvents and reagents used were of analytical grade.

2.2. Endothelial cell culture

EC were obtained from the American Type Culture Collection (Cows Pulmonary Artery Endothelium, CPAE, CCL-209) and used between passage 18 and 26 population doublings. Cells were grown in Minimum Essential Medium (MEM, with Earle's Salts; Gibco, Life Technologies) containing 20% FBS (Performance Plus, United States, Gibco, Life Technologies), 1% NEAA (Non-Essential Amino Acid solution 100X, Lonza), and 1% Pen-Strep (10,000 units/mL penicillin and 10,000 μ g/mL streptomycin, Gibco, Life Technologies) at 37 °C in a fully humidified atmosphere of 5% CO₂.

2.3. Endothelial cell treatment protocols

Sub-confluent EC were seeded in six-well plates 12 h before each experiment. Egt (Santa Cruz Biotechnology, Santa Cruz, CA) (up to 1 mM) was added 12 h before starting the short-term (48 h, 72 h) exposure to high-glucose (hGluc) (25 mM) in complete culture medium (Egt+hGluc). The effect of Egt itself was evaluated by treating cells with Egt alone (Egt) in medium containing normal-glucose (nGluc) concentration (5.5 mM). Control cells (Ctrl) were incubated only with nGluc. Osmotic control was performed by treatment with 25 mM L-glucose.

2.4. SIRT1 activity inhibition

For SIRT1 inhibition, sirtinol (Srt) (50 μ M dissolved in DMSO 0.02%) was added to EC 3 h before treatment with hGluc (25 mM) or nGluc (5.5 mM) in the presence or absence of Egt (0.5 mM). Control cells (Ctrl) were incubated with nGluc containing 0.02% DMSO.

2.5. SIRT6 gene silencing

EC were transfected with small interfering RNA (siRNA) (30 nM) and with control non targeting siRNA (NT-siRNA) (30 nM) using RNAiFectin™ transfection reagent SIRT6 pool siRNA consisted of a mixture of three sequences designed for specific Bos taurus Sirtuin 6 (NCBI reference sequences: NM_001098084.1). Transfection was performed following the manufacturer's instructions. Briefly, EC (3×10^5 cells/well) were seeded in a six well tissue culture plate 24 h prior to transfection in complete culture medium and maintained at 37 °C in 5% CO₂. After removing the growth medium, the transfection complexes (siRNA-RNAiFectin™) were added to serum-free and antibiotic-free medium. Cells were

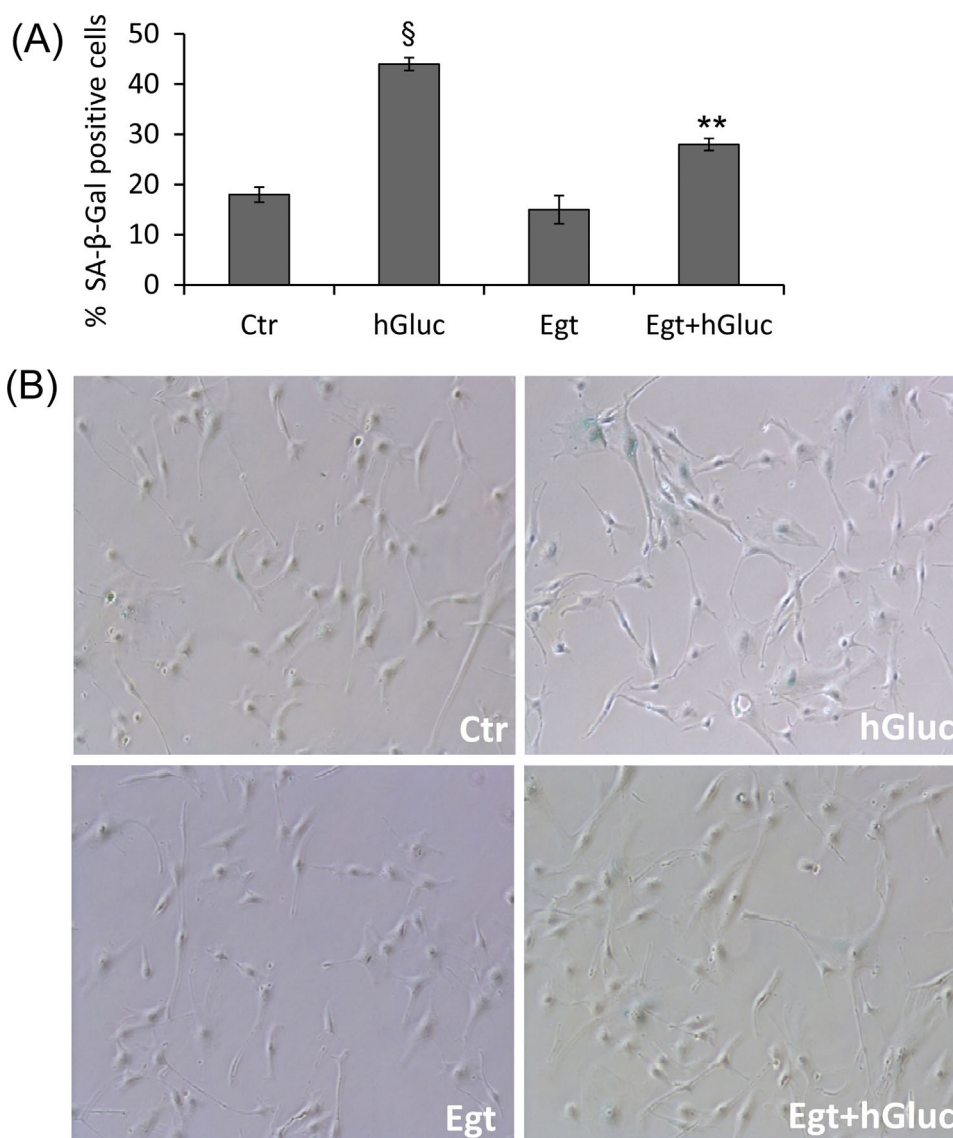


Fig. 3. Effect of ergothioneine on high-glucose induced EC senescence. (A) Percentage of SA-β-galactosidase staining positive EC grown for 48 h with high-glucose (hGluc), high-glucose plus ergothioneine (0.5 mM) (Egt+hGluc), ergothioneine (Egt) (0.5 mM), or media containing nGluc (Ctr). (B) Representative images of SA-β-galactosidase staining (blue). Data are the mean $\pm$ SD of five independent experiments with ** $P < 0.01$ vs. hGluc and § $P < 0.01$ vs. Ctr. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

incubated for 8 h, followed by additional 16 h of incubation after the addition of FBS (10%) directly to each well. After incubation at 37 °C in 5% CO₂ for a total of 24 h, EC were used for treatments.

2.6. Cell viability assay

Cell viability was determined by XTT (sodium 30-[1-(phenylaminocarbonyl)-3,4-tetrazolium]-bis (4-methoxy-6-nitro) benzene sulfonic acid hydrate) assay (Roche) following manufacture's protocol. Absorbance rate was measured at 490 nm using a micro plate reader (BIO-RAD). Cell growth data were expressed as percentage of the control.

2.7. Intracellular ROS detection

Intracellular ROS were detected by means of the oxidation-sensitive fluorescent probe 2',7'-dichlorofluorescein diacetate (DCFH-DA, Sigma-Aldrich Co., St. Louis, MO, USA) following manufacture's protocol. Briefly, EC were grown in a six-well plates, pre-incubated with DCFH-DA for 30 min before treatment with

hGluc alone and in the presence of EGT. After incubation, the media was removed and cells were washed twice with 1X PBS. EC lysates were prepared by adding Tris-HCl 0.5 M, pH 7.6, 1% SDS. The fluorescence intensity of 2',7'-dichlorofluorescein (DCF) was quantified on a Perkin Elmer Life Sciences LS 55 spectrofluorometer using an excitation wavelength of 488 nm and an emission wavelength of 530 nm. The fluorescence intensity, corrected by subtracting the control fluorescence value (F-F0) was expressed as relative fluorescence units (RFU).

2.8. Mass spectrometric analysis of cellular ergothioneine oxidation products

With the aim to exclusively analyze the cellular products of the ergothioneine oxidation, EC (2×10^6 cells/plate) were pre-incubated with Egt (0.5 mM) for 12 h and, then, the pre-incubation media were removed before hGluc exposure. This procedure allowed us to avoid any chemical interference due to the presence of Egt in the media throughout the hGluc exposure. To this end, after pre-incubation, EC were washed twice with phosphate-buffered

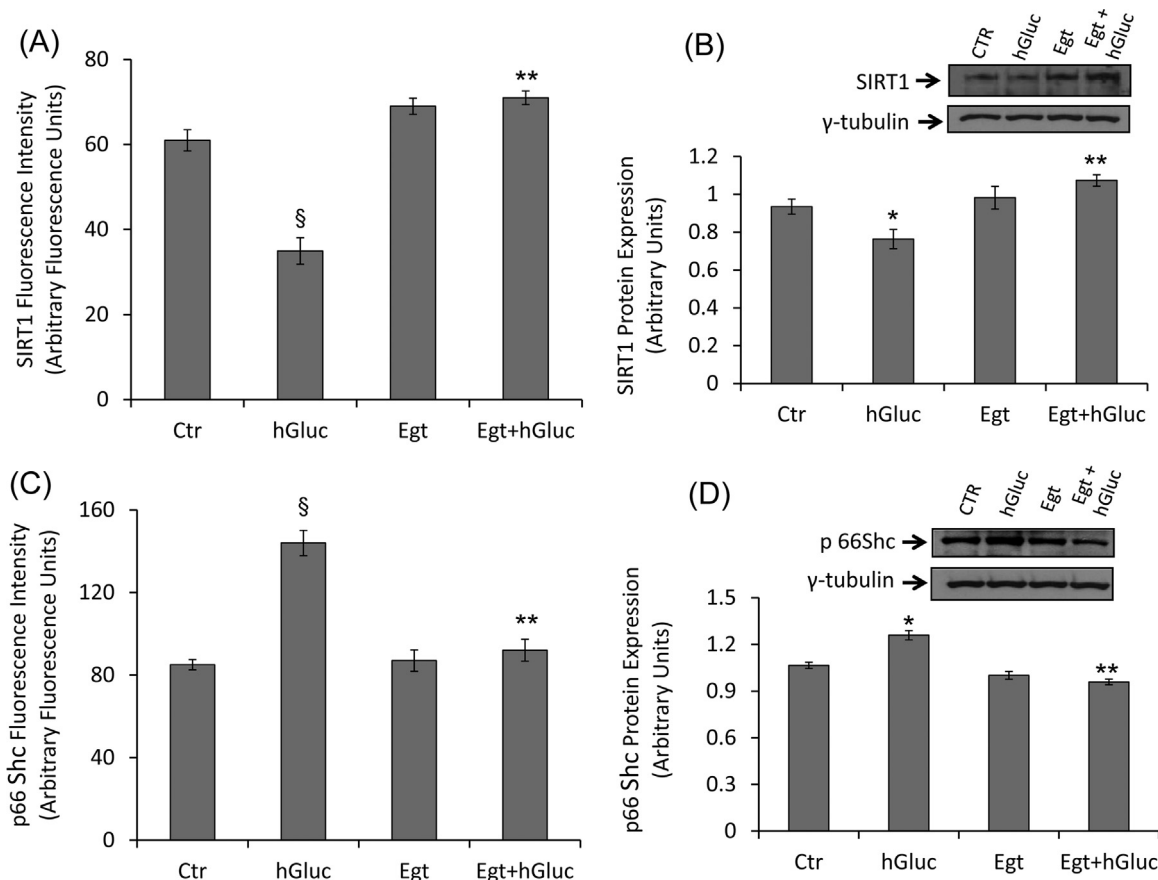


Fig. 4. SIRT1 and p66shc regulation by ergothioneine. (A, C) Fluorescence intensity determined by confocal laser scanning microscopy analysis and (B, D) Western blot analysis of protein levels of SIRT1 and p66shc in EC pre-treated for 12 h with Egt and then subjected for 48 h to high-glucose (hGluc), high-glucose plus ergothioneine (Egt + hGluc), ergothioneine (Egt) alone, or media containing nGluc (Ctrl). The Egt concentration used was 0.5 mM. The fluorescence intensity was calculated with Image J software analysis and expressed as arbitrary fluorescent units (AFU). The mean of the unspecific fluorescence of the secondary antibodies was calculated with Image J software and subtracted. Data are mean $\pm$ SD of six independent experiments with ^{**} $P < 0.01$ vs. hGluc, ^{*} $P < 0.05$ vs. Ctrl, and [§] $P < 0.01$ vs. Ctrl.

saline (PBS) solution before performing the hGluc incubation as described above. After hGluc exposure, cells were washed twice with PBS and scraped in cold methanol (800 μ L). The scraped cells were transferred into eppendorf tubes, sonicated, and centrifuged at 3800g for 10 min. The supernatants were collected and evaporated to dryness by UNIVAPO 100 H (Uniequip). Subsequently, each sample was resuspended with 200 μ L of 0.1% formic acid. Sample volumes of 20 μ L were injected for the mass spectrometric analyses performed, as previously described [21], with an HPLC Agilent 1100 series coupled on line with an Agilent LC-MSD SL quadrupole ion trap. Briefly, all MS acquisitions were performed by using electrospray ionization (ESI) in positive ion mode Nitrogen was used as a drying and a nebulising gas at a flow rate of 7 L/min and a pressure of 30 psi and the nebulizer temperature was set at 350 °C. The ion charge control (ICC) was applied with a target set at 30,000 and maximum accumulation time at 20 ms. The measurements were performed from the peak area of the extracted ion chromatogram (EIC) and compared with the standard solutions. The optimization of the instrumental parameters for the analyses of Egt and EH was performed by continuous infusion (FIA-ESI-MS/MS measurements) of 5 μ M standard solution in 0.1% formic acid. The mass cutoff and amplitude were optimized to obtain the most efficient MS/MS transitions from the positively charged precursor ion $[M+H]^+$ to the fragment ions. Multiple reaction monitoring (MRM) was used for the detection of analytes in HPLC-ESI-MS/MS analyses. The most intense transitions utilized for MS/MS detection were 230.1 $\rightarrow$ 186 for Egt and 198.1 $\rightarrow$ 154 for EH. The chromatographic analyses with ESI-MS detection were performed on a

250 mm $\times$ 3.0 mm i.d., 5 μ m, Discovery-C8 analytical column (Supelco, PA, USA). The mobile phase was composed with 0.1% formic acid in water and pumped at 0.1 mL/min in isocratic conditions at room temperature.

2.9. Western blot analysis

EC monolayers were washed twice with 1 $\times$ PBS and immediately scraped off the plate in lysis buffer (50 mM Tris, pH 7.4, 150 mM sodium chloride, 1% Nonidet P-40, 1 mM EDTA, 1 mM sodium orthovanate, 1 mM sodium fluoride, 1 μ g/mL leupeptin, 1 μ g/mL aprotinin, 1 μ g/mL pepstatin A, 1 mM PMSF). Scraped cells were transferred to a microcentrifuge tube, kept on ice, and sonicated for 10–15 s to complete cell lysis. After 30 min of gently shaking at 4 °C, lysates were centrifuged at 10,000g for 10 min and supernatants were collected. Approximately, 30 μ g of protein extracts were separated by 10% SDS-polyacrylamide gel electrophoresis and transferred onto nitrocellulose membranes by Trans-Blot Turbo Transfer System (BioRad). Membranes were blocked in 5% w/v milk, 1X TBS, 0.1% Tween-20 at 25 °C for 2 h with gentle shaking, and then incubated overnight at 4 °C with antibodies against SIRT1 (1:3000) (rabbit monoclonal, ab32441, Abcam, Cambridge, UK), SIRT6 (1:1000) (rabbit polyclonal, ab62738, Abcam, Cambridge, UK), NF- κ B (1:1000) (rabbit monoclonal, C22B4, Cell Signaling Technology, Danvers, MA), and against gamma-tubulin (1:1000) (mouse monoclonal, GTU-88, Sigma, St. Louis, MO, USA). After incubation with secondary antibodies (1:10,000), membranes were washed three times and bands were detected by

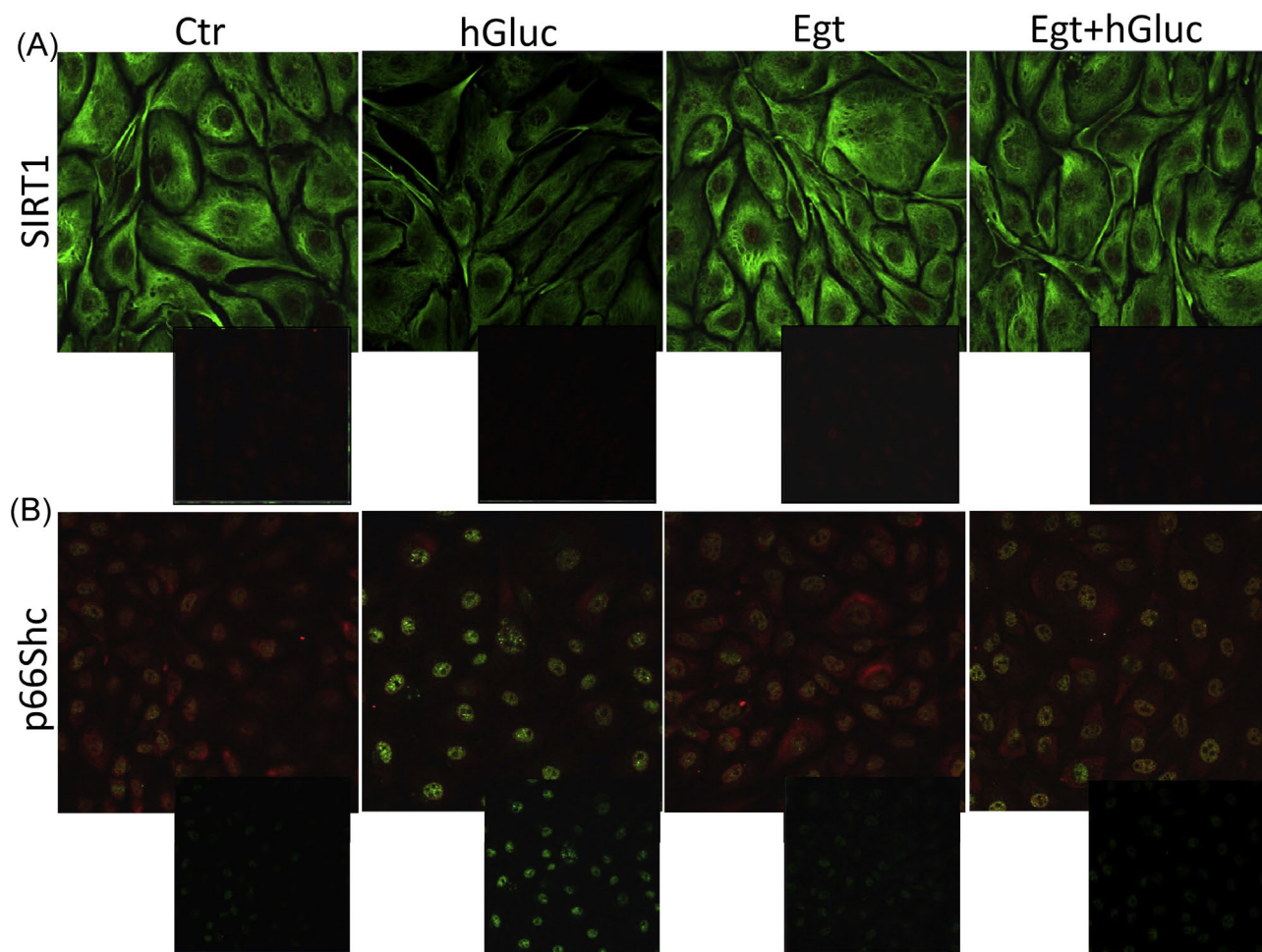


Fig. 5. Representative confocal images of SIRT1 and p66shc expression in EC. (A) Merge immunofluorescence images of SIRT1 (red)/vimentin (green) and (B) p66shc (green)/vimentin (red); (insets) respective images of SIRT1 (red) and p66^{shc} (green) fluorescence intensities. Secondary antibodies were Alexa Fluor[®] 488 or Alexa Fluor[®] 633. The Egt concentration used was 0.5 mM. Data are representative of six independent experiments. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

the enhanced chemiluminescence kit (Immobilon Western, Chemiluminescent HRP Substrate, Millipore, Billerica, MA 01821, USA). Semiquantitative densitometry of Western blots was performed using a Scan LKB (Amersham Pharmacia).

2.10. Confocal laser scanning microscopy

EC cultured in 24-well plates containing microscope glass (12 mm) (Thermo Scientific) were fixed with 3% (v/v) paraformaldehyde for 20 min. and permeabilized with 0.1% (v/v) Triton X-100 in PBS for 10 min at room temperature. Immunofluorescence detection of SIRT1, SIRT6 and NF- κ B was performed by using specific antibodies against Vimentin (1:1,000, mouse) (Sigma, St. Louis, MO,) SIRT1 (1:100) (rabbit monoclonal, ab32441, Abcam, Cambridge, UK), SIRT6 (1:500) (rabbit polyclonal, ab62738, Abcam, Cambridge, UK), and against NF- κ B (1:100) (rabbit monoclonal, C22B4, Cell Signaling Technology, Danvers, MA). Secondary antibodies were Alexa Fluor[®] 488 (1:1,000) or Alexa Fluor[®] 633 (1:1,000) (Life Technologies Italia, Monza, Italy). Microscopy analyses were performed using Zeiss LSM 700 confocal microscope equipped with a plan apochromat X63 (NA 1.4) oil immersion objective. The fluorescence intensity was evaluated with ImageJ software and expressed as arbitrary fluorescence units (AFU).

2.11. β -Galactosidase activity assay

Senescence-associated β -galactosidase (SA- β -Gal) activity was determined by senescence cells histochemical staining kit following manufacturer's procedure (Cell Signalling Technology Danvers, MA). Briefly, stained cells were viewed under a Nikon Eclipse TE300 Inverted microscope (Nikon Instruments, Inc., Melville, NY). Results were expressed as the percentage of SA- β -galactosidase positive cells/field.

2.12. Statistical analysis

The data shown were mean values of at least four independent experiments and expressed as mean $\pm$ SD. Statistical analysis was performed by the Student's *t* test. Statistical significance was set at a level of $P < 0.05$.

3. Results

3.1. Cytoprotective and antioxidant effects of ergothioneine during high-glucose treatment

The cell viability assay was used to investigate whether increasing concentrations of Egt alone could cause reduced cell viability and to evaluate the efficacy of Egt in protecting EC against

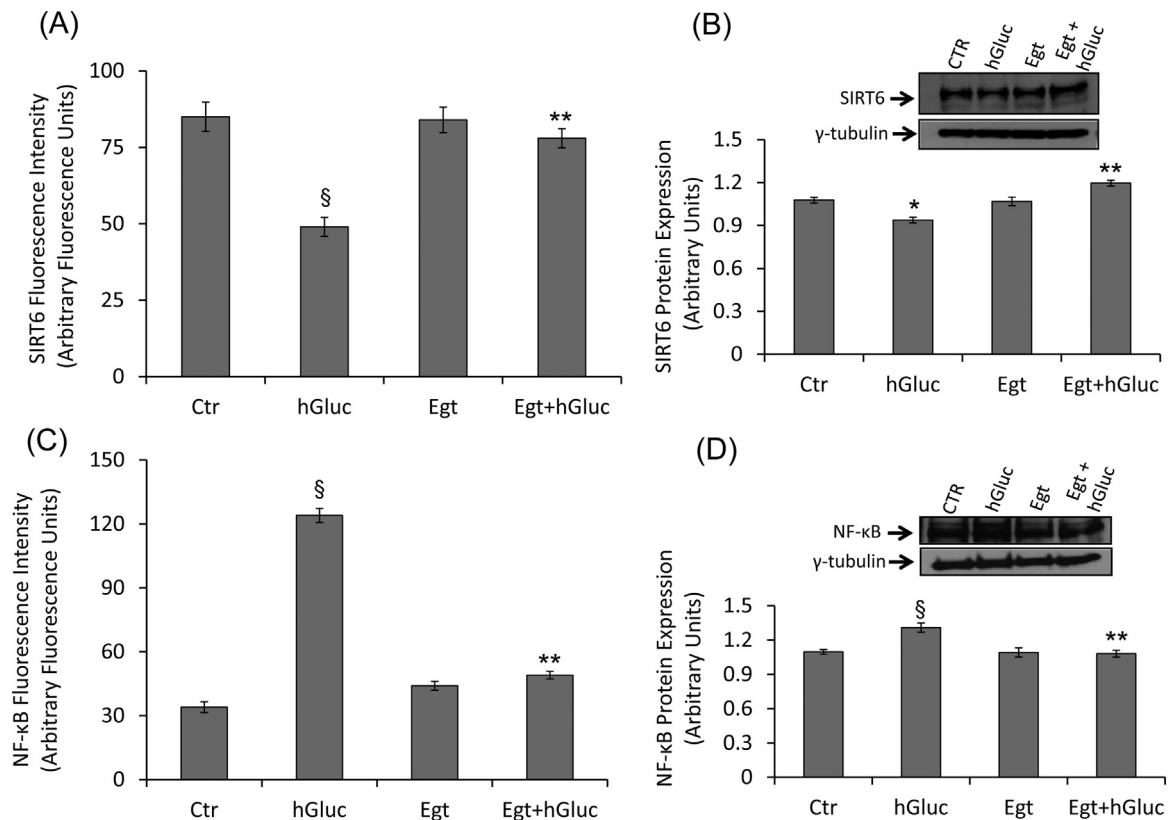


Fig. 6. Effect of ergothioneine on SIRT6 and NF-κB signaling. Expression levels of SIRT6 and NF-κB were determined by (A, C) confocal laser scanning microscopy and (B, D) Western blot analysis. EC were pre-treated for 12 h with Egt followed by 48 h treatment with high-glucose (hGluc), high-glucose plus ergothioneine (Egt+hGluc), ergothioneine (Erg) alone, or media containing nGluc (Ctrl). The Egt concentration used was 0.5 mM. The fluorescence intensity, expressed as arbitrary fluorescent units (AFU), was calculated with Image J software analysis. The mean of the unspecific fluorescence of the secondary antibodies was calculated with Image J software and subtracted. Data are mean $\pm$ SD of six independent experiments with ^{**} $P < 0.01$ vs. hGluc, ^{*} $P < 0.05$ vs. Ctrl, and [§] $P < 0.01$ vs. Ctrl.

hGluc-induced cytotoxicity. Thus, EC were exposed to increasing concentration of Egt (0.01, 0.025, 0.05, 0.1, 0.25, 0.5, and 1 mM) for 48 h. In accordance with previous studies [22,24], the dose-response results indicated that Egt does not alter cell viability up to 1 mM concentration (Fig. 2A). Notably, co-treatment of Egt and hGluc (Egt+hGluc) reduced the cytotoxic effect of hGluc in a dose-dependent manner, with highest potency starting at 0.1 mM ($P < 0.01$ vs. hGluc). At the Egt concentration of 0.5 mM, chosen for our studies, the cytotoxic effect of hGluc was consistently decreased. Specifically, after 48 h treatment with Egt+hGluc, the cell viability was 1.28 ± 0.021 O.D. compared with 0.811 ± 0.03 O.D. of cells treated with hGluc only ($P < 0.01$) (Fig. 2A). This effect lasted up to 72 h, even if the cell viability was improved to a less extent ($P < 0.05$ vs. hGluc) (Fig. 2B). DCFH-DA assay for ROS indicated that the cytoprotective effect of Egt (0.5 mM) was paralleled by its antioxidant effect. Indeed, the hGluc-increased fluorescence (512 ± 33 in hGluc vs. 235 ± 10 in Ctrl) was reduced by Egt (280 ± 29 in Egt+hGluc vs. 512 ± 33 in hGluc) ($P < 0.01$ vs. hGluc) (Fig. 2C).

3.2. Ergothioneine effect on endothelial cell senescence

The increased cell senescence is a critical event during hyperglycaemia-induced endothelial dysfunction [1,14,32]. Accordingly, the percentage of senescent EC reached a significant level after 48 h of treatment with hGluc ($44.2 \pm 1.31\%$ vs. $18.1 \pm 1.52\%$ in Ctrl, $P < 0.01$) (Fig. 3A). SA β -gal positive cells in samples co-treated with Egt+hGluc were $28.3 \pm 1.22\%$ vs. $44.2 \pm 1.31\%$ in hGluc ($P < 0.01$). Moreover, microscopy examination revealed that the hGluc senescent EC become large and with more spindle-shaped morphology (Fig. 3B). The Egt treatment in the presence of hGluc

(Egt+hGluc) resulted in a morphological adjustment to the control cell-like small spindle shape. Finally, the morphological similarity between control (Ctrl) and Egt samples (Egt) supports the absence of cytotoxicity by Egt itself (Fig. 3B).

3.3. Ergothioneine effect on SIRT1 and p66Shc protein expression

When we evaluated the possible effect of Egt on the down-regulation of SIRT1 signaling associated with hGluc toxicity, both Western blot and confocal laser-scanning microscopy analyses revealed that Egt modulates SIRT1 expression levels in Egt+hGluc samples (Figs. 4A, B and 5A). Specifically, SIRT1 arbitrary fluorescence units (AFU) were significantly decreased by 48 h treatment with hGluc (35.2 ± 3.1 vs. 61.2 ± 2.5 AFU in control cells, $P < 0.01$) whereas Egt+hGluc cells showed a SIRT1 fluorescence intensity significantly higher than that observed in hGluc cells (71 ± 1.6 vs. 35.2 ± 3.1 AFU in hGluc, $P < 0.01$) (Figs. 4A and 5A). The Egt efficiency in the modulation of SIRT1 was also confirmed by Western Blot analysis (Fig. 4B).

SIRT1 protects blood vessels from hyperglycemia-induced endothelial dysfunction through a mechanism involving the down-regulation of the adapter protein p66Shc expression, which directly stimulates mitochondrial ROS generation by antioxidant activity and is capable of promoting a redox reaction with cytochrome c, leading to oxygen reduction and H_2O_2 formation [33]. Our results indicated that the effect of hGluc on the downregulation of SIRT1 was paralleled by the upregulation of p66Shc protein expression (144 ± 6.1 AFU vs. 85 ± 2.5 AFU in control cells $P < 0.01$) and, notably, that Egt+hGluc treatment resulted in the downregulation of p66Shc (92 ± 5.1 AFU vs. 144 ± 6.1 AFU in hGluc $P < 0.01$) (Figs. 4C, D and 5B).

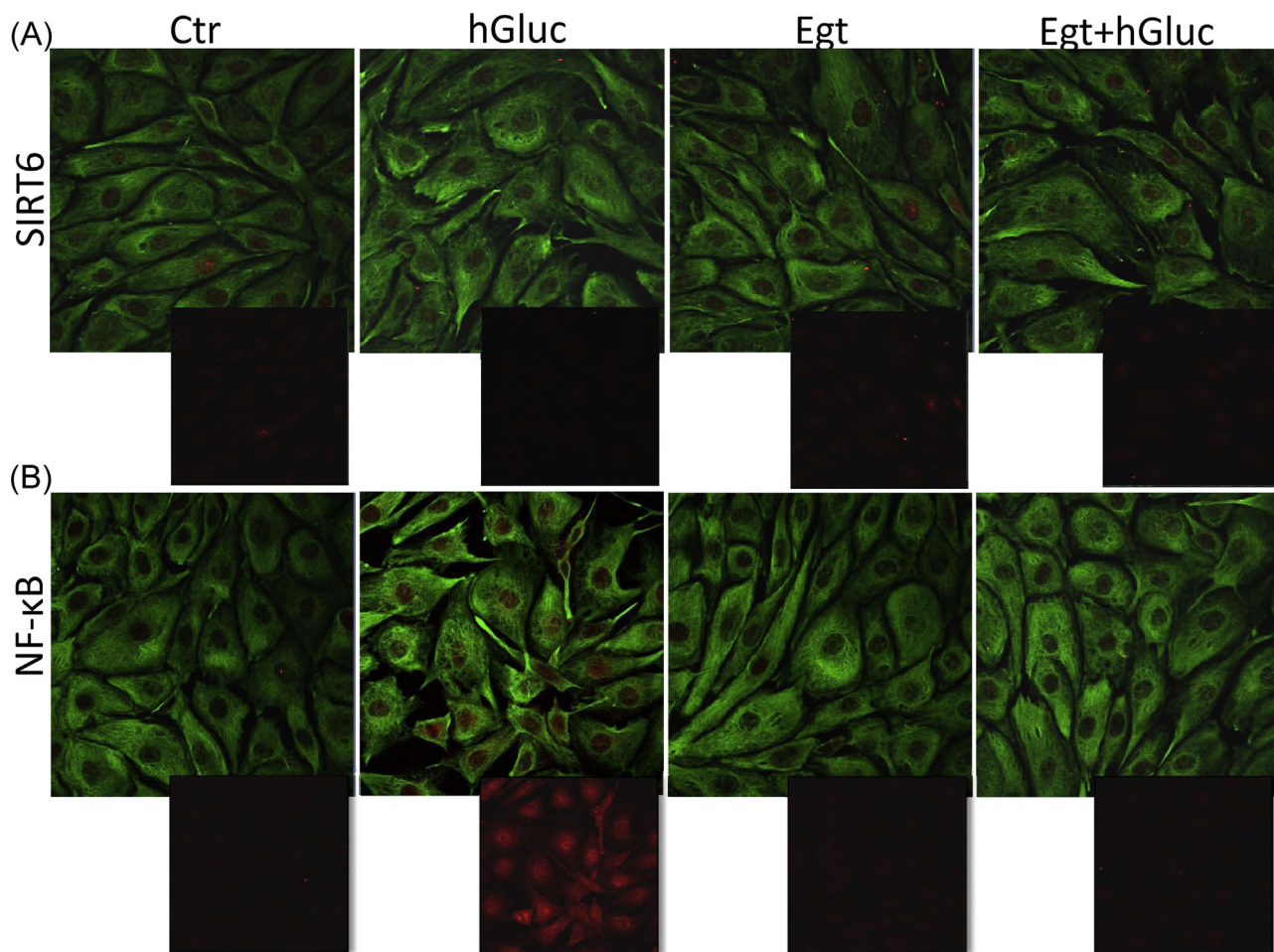


Fig. 7. Representative confocal images of SIRT6 and NF-κB expression in EC. (A) Immunofluorescence analysis of SIRT6 (red) and vimentin (green), and (B) NF-κB (red) and vimentin (green); (insets) corresponding images of SIRT6 (red) and NF-κB (red) fluorescence intensities. The Egt concentration used was 0.5 mM. Secondary antibodies were Alexa Fluor[®] 488 or Alexa Fluor[®] 633. Data are representative of six independent experiments. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

3.4. Ergothioneine effect on SIRT6 and NF-κB protein levels

Endothelial dysfunction and senescence occur *via* SIRT6 and NF-κB signaling, shown to be finely regulated when EC are exposed to hGluc and hydrogen peroxide [9,34–36]. SIRT6 plays a key role in stress response by the transcriptional repression of NF-κB-dependent targets and, specifically, the interaction with NF-κB-signalling contributes to premature senescence. Indeed, SIRT6 deficiency in EC increased the expression of pro-inflammatory cytokines such as IL-1β along with an increased transcriptional activity of NF-κB [35]. Moreover, the hGluc induced loss of SIRT6 was associated with an increased expression of NF-κB, whereas overexpression of SIRT6 and treatment with positive modulators of SIRT6 were associated with decreased NF-κB transcriptional activity [9]. Accordingly, results indicated that short-term exposure of EC to hGluc concentration downregulates SIRT6 protein (49.2 ± 3.1 AFU vs. 85 ± 4.8 AFU in control cells, $P < 0.01$) with a concomitant upregulation of NF-κB protein expression (124 ± 3.3 AFU vs. 34 ± 2.5 AFU in control cells, $P < 0.01$) (Figs. 6 and 7). These effects were blocked by Egt ($P < 0.01$) and, more in details, SIRT6 expression in Egt+hGluc cells was 78.3 ± 3.1 AFU vs. 49.2 ± 3.1 AFU in hGluc cells ($P < 0.01$) (Figs. 6A and 7A), whereas NF-κB expression was 48.9 ± 1.8 AFU vs. 124 ± 3.3 AFU in hGluc cells ($P < 0.01$) (Figs. 6C and 7B). Results of Western blot analysis also indicated that hGluc downregulates SIRT6 ($P < 0.05$ vs. Ctr) and upregulates NF-κB protein expression ($P < 0.01$ vs. Ctr) (Fig. 6B, D).

Moreover, a reduced SIRT6 downregulation (Fig. 6B) and a reduced NF-κB upregulation ($P < 0.01$ vs. hGluc) (Fig. 6D) were observed during hGluc treatment in the presence of Egt (Egt+hGluc).

3.5. Inhibition of SIRT1 activity by sirtinol

In order to demonstrate whether the protective effect of Egt was SIRT1-dependent, the effect of Egt was investigated in sirtinol (Srt)-treated EC, a SIRT1 inhibitor, during hGluc and nGluc exposure. Under nGluc conditions, Srt increased cell senescence of about 3.6-fold ($P < 0.01$ vs. Ctr) and this effect continued in the presence of Egt (Egt+Srt) ($P < 0.01$) (Fig. 8A, B). In EC exposed to hGluc, cell senescence decreased in the presence of Egt treatment (Egt) ($P < 0.01$) but not when Egt was used in the presence of Srt (Egt+Srt) (Fig. 8C, D). These results indicated that Egt-induced reduction of endothelial senescence following hGluc exposure was mediated by a SIRT1-dependent pathway.

3.6. SIRT6 silencing

To investigate the role of SIRT6 in the pathway of Egt protection against hGluc-induced senescence, the *SIRT6* gene was silenced using a specifically designed pool of siRNA. Reduction of SIRT6 protein levels in silenced cells after 24 h of transfection was confirmed by Western blot analysis (Fig. 9A, B). After 24 h of transfection, when cells were cultured for additional 48 h with nGluc

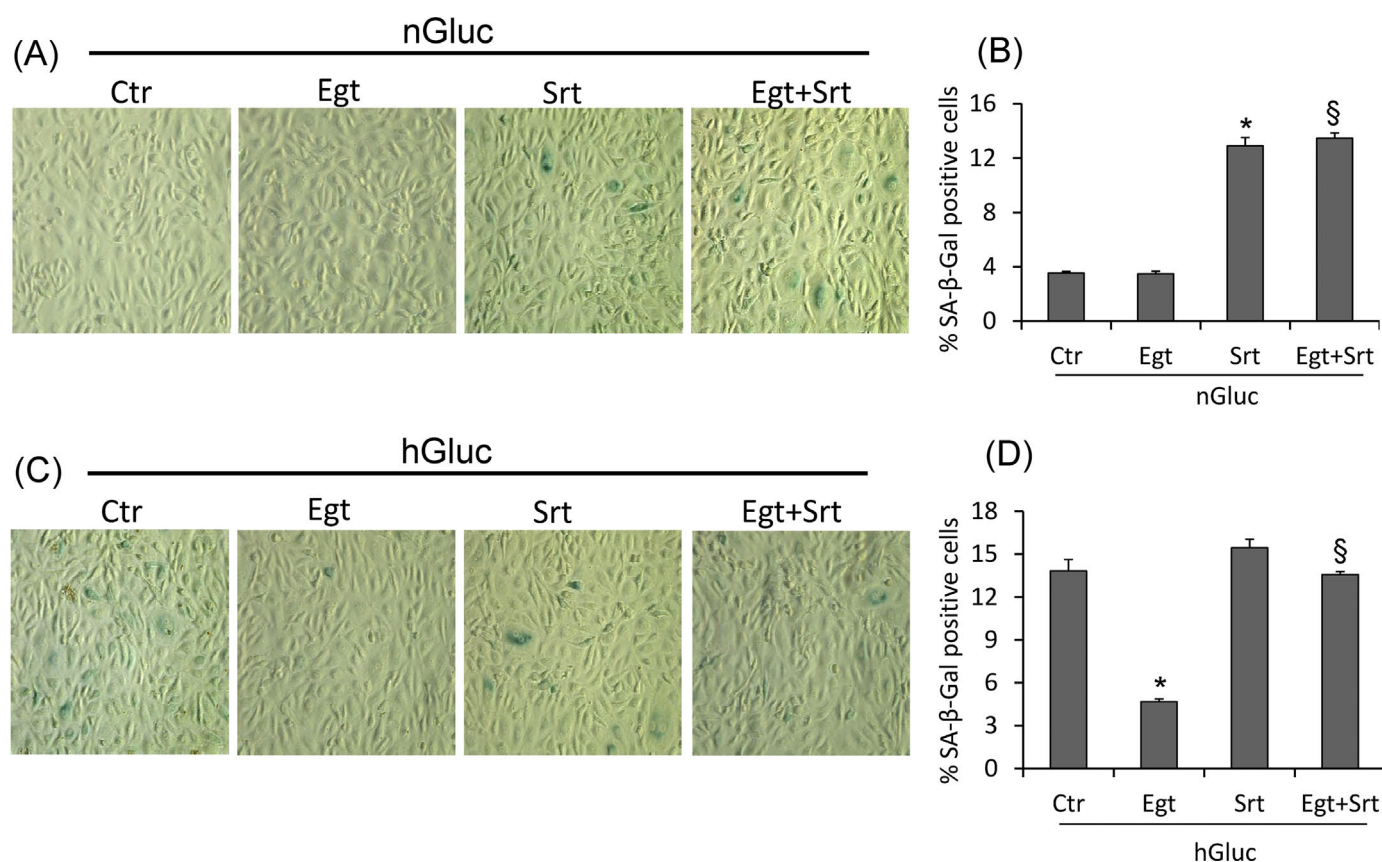


Fig. 8. Effect of SIRT1 inhibition on the ergothioneine protection against high-glucose induced EC senescence. Sirtinol (Srt) (50 μ M dissolved in DMSO 0.02%) was added to EC 3 h before treatment for 48 h with (A, B) normal-glucose (nGluc) (5.5 mM) or (C, D) high-glucose (hGluc) (25 mM), alone or combined with Egt (0.5 mM). Control cells (Ctr) were incubated with medium containing 0.02% DMSO. Representative images of SA- β -galactosidase staining (blue). Values represent the mean $\pm$ SD of four independent experiments with * P < 0.01 vs. Ctr and § P < 0.01 vs. Egt. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

alone or in the presence of Egt, results indicated that the cellular senescence was consistently higher in SIRT6-siRNA treated cells, either in SIRT6-siRNA-nGluc (P < 0.01 vs. Ctr-siRNA-nGluc) and in SIRT6-siRNA-Egt + nGluc (P < 0.01 vs. Ctr-siRNA-Egt + nGluc) (Fig. 9C, D). Under nGluc conditions, comparison between SIRT6-siRNA and Ctr-siRNA transfected cells demonstrated that SIRT6 is crucial in protecting EC against senescence. In cells treated with hGluc for 48 h after transfection, SIRT6-silenced EC (SIRT6-siRNA-hGluc) displayed lower protein expression levels compared to Ctr-siRNA-hGluc (P < 0.05) (Fig. 9E, F) and, importantly, the Egt was not more effective in reducing the hGluc-induced endothelial senescence (SIRT6-siRNA-Egt + hGluc vs. SIRT6-siRNA-hGluc) (Fig. 9G, H), suggesting that this compound acts through a SIRT6-dependent mechanism(s).

3.7. Ergothioneine oxidation following high-glucose exposure

The Egt chemical antioxidant mechanism involves the production of EH, the main Egt oxidation product found in human neutrophils subjected to phorbol 12-myristate 13-acetate-induced oxidative burst and in cell-free oxidation reactions with hypochlorite, hydrogen peroxide, and peroxynitrite [21]. Here, the HPLC-ESI-MS analyses of the cellular Egt uptake and oxidation revealed a consistent increase of the intracellular levels of Egt during the pre-incubation time preceding the short-term exposure to hGluc (P < 0.01 vs. time zero) (Fig. 10A). Results indicated that nGluc treatment did not significantly alter the cellular levels of Egt and EH (Fig. 10B). On the contrary, hGluc determined a noticeable time-dependent decrease of the intracellular Egt levels (P < 0.05

vs. nGluc), accompanied by an increased production of EH (P < 0.05 vs. nGluc) (Fig. 10B).

4. Discussion

This study reports the first demonstration that the protective role of Egt against the hGluc-induced endothelial senescence is exerted through its uncommon redox mechanism and via the regulation of SIRT1/SIRT6 expression and their cellular targets, NF- κ B and p66Shc.

As antioxidant molecule, Egt has received limited attention regarding its potential role in counteracting oxidative stress in CVD, although vascular cell studies show that the extracellular thiol/disulfide redox state plays a vital role in controlling CVD through proinflammatory signaling [37]. Indeed, both EC and monocyte respond to oxidized cysteine with increased cell adhesion and proinflammatory signaling [37]. Similarly to alkylthiols, such as cysteine or glutathione, Egt has been considered to oscillate between the reduced state and the oxidized state represented by the disulfide form (ESSE) [38,39]. Probably, this reason limited the view of the role of Egt as cellular antioxidant because it was not clear the advantage with respect to alkylthiols, which appear better reducing agents if one compares the respective reversible redox potentials for the equilibrium sulfide/disulfide. However, our recent findings reveal the uncommon redox behavior of Egt, based on the irreversible transformation of intermediate oxidation products, which makes Egt an antioxidant absolutely different from alkylthiols and opens new perspectives on its cellular role

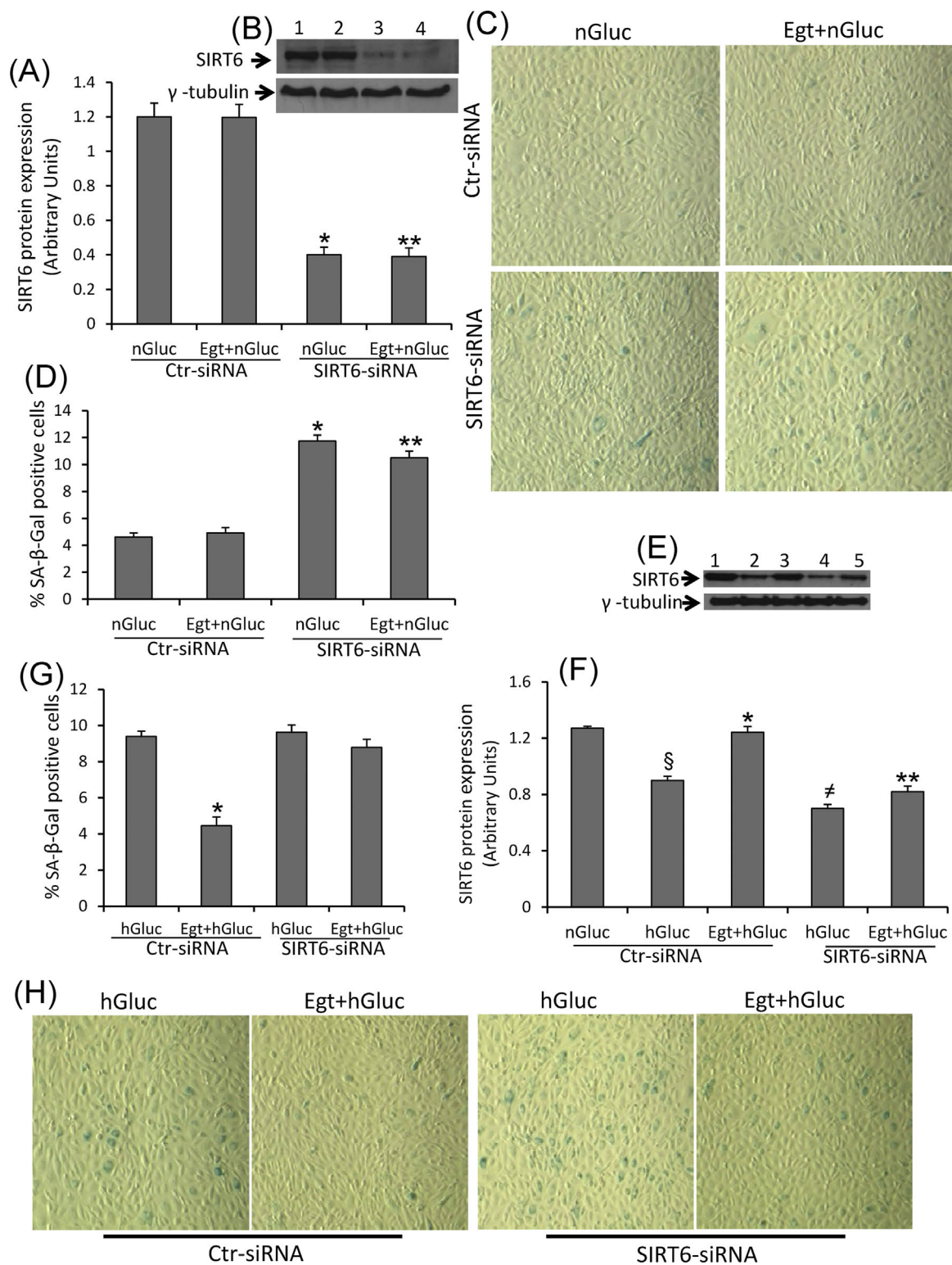


Fig. 9. Effect of SIRT6 silencing on the beneficial effect of ergothioneine during high-glucose exposure. EC were transfected with SIRT6-siRNA (30 nM) or with control NT-siRNA (Ctr-siRNA) (30 nM) for 24 h following the manufacturer's instructions. After transfection, cells were treated with normal-glucose (nGluc) (5.5 mM) or high-glucose (hGluc) (25 mM), either alone or combined with Egt (0.5 mM) for 48 h. (A, B) Western blot analysis of SIRT6 in transfected EC under nGluc treatment; lane 1 = Ctr-siRNA-nGluc; lane 2 = Ctr-siRNA-Egt + nGluc; lane 3 = SIRT6-siRNA-nGluc; lane 4 = SIRT6-siRNA-Egt + nGluc. * $P < 0.01$ vs. Ctr-siRNA-nGluc, ** $P < 0.01$ vs. Ctr-siRNA-Egt + nGluc. (C, D) Percentage and representative images of SA-β-galactosidase staining positive cells (blue). * $P < 0.01$ vs. Ctr-siRNA-nGluc, ** $P < 0.01$ vs. Ctr-siRNA-Egt + nGluc. (E, F) Western blot analysis of SIRT6 in transfected EC under nGluc and hGluc treatments; lane 1 = Ctr-siRNA-nGluc; lane 2 = Ctr-siRNA-hGluc; lane 3 = Ctr-siRNA-Egt + hGluc; lane 4 = SIRT6-siRNA-hGluc; lane 5 = SIRT6-siRNA-Egt + hGluc. § $P < 0.01$ vs. Ctr-siRNA-nGluc, * $P < 0.01$ vs. Ctr-siRNA-hGluc, § $P < 0.05$ vs. Ctr-siRNA-hGluc, ** $P < 0.05$ vs. SIRT6-siRNA-hGluc. (G, H) SA-β-galactosidase staining assay of SIRT6 silenced EC under hGluc treatment, * $P < 0.01$ vs. Ctr-siRNA-hGluc. Values are the mean $\pm$ SD of three independent experiments. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

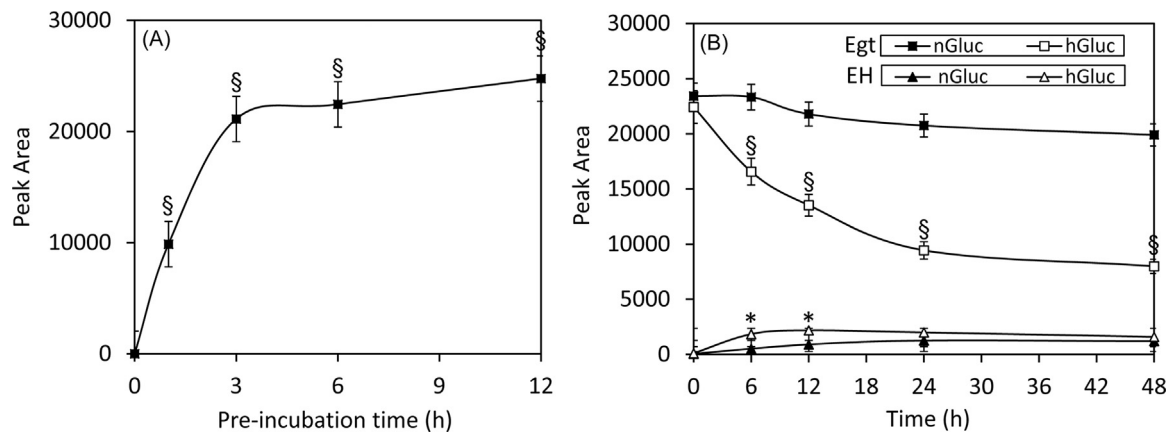


Fig. 10. Time course of ergothioneine uptake and oxidation. (A) EC were incubated up to 12 h with Egt (0.5 mM). After media removal, cells were prepared for HPLC-ESI-MS² analysis as described under Section 2. Peak area, determined following the MS² transition 230.1 → 186, are mean ± SD of four independent experiments with [§]*P* < 0.01 vs. time zero (B) After 12 h pre-incubation with Egt (0.5 mM), the culture media was removed and cells were washed PBS solution. The time-dependent exposure to hGluc was performed up to 48 h and, then, samples were prepared for HPLC-ESI-MS² analysis of the cellular levels of Egt (■), following the MS² transition 230.1 → 186, and EH (▲) production, following the MS² transition 198.1 → 154). Data are mean ± SD of four independent experiments with **P* < 0.05 vs. respective nGluc, and [§]*P* < 0.01 vs. respective nGluc.

[21]. Accordingly, this study shows that the intracellular levels of EH increase during the short-term exposure to hGluc whilst Egt is consumed. The vascular protective effect of Egt occurs through reduction of ferrylmyoglobin and the modulation of heat shock protein 70 and proinflammatory cytokines (TNF- α and IL-1 β) [26]. In addition, iron and copper chelators can be protective by decreasing the production of free radical during ischemia/reperfusion, suggesting another mechanism by which Egt may protect [27,40,41]. Moreover, the antioxidant effect of Egt on human brain microvascular EC, which express the specific transporter OCTN-1, indicated that this compound prevents endothelial dysfunction under oxidative stress conditions, such as diabetes [22].

In humans, Egt accumulates in various cells and tissues, including erythrocytes, bone marrow, liver, kidney, eye cornea, eye lens, and seminal, at high concentrations from 100 μ M to 2 mM [19]. Fluorescence detection of human plasma Egt by high-performance liquid chromatography and capillary electrophoresis in 34 volunteers showed a concentration of about 1.71 μ M/L [42]. In our study, Egt did not affect EC viability from μ M to mM concentrations. Noticeable, starting from 0.1 mM, Egt blocked the detrimental effect of hGluc on cell viability and senescence, with the highest trend at 0.5 mM. The mM range of concentration has been shown to be also effective in reducing adhesion molecule expression in human aortic EC and the hGluc induced oxidative stress and cytotoxicity in human brain microvascular EC [22,24].

SIRT1 and SIRT6, regarded as lifespan-controlling proteins, are negative regulators of NF- κ B and p66Shc [34,35,43–46]. SIRT1 exerts atheroprotective effect through the activation of eNOS or the inhibition of NF- κ B activity in macrophages and endothelial cells [11]. SIRT6 controls the inflammatory pathway(s) in diabetic atherosclerotic lesions and, at endothelial level, its expression is reduced following treatment with hydrogen peroxide [36]. Moreover, SIRT6 downregulation during short-term exposure to hGluc is paralleled by the increased expression of NF- κ B whereas overexpression of SIRT6 in the presence of incretin treatment relates to a decreased NF- κ B expression [9].

In EC exposed to hGluc and in aortas of diabetic mice, mitochondrial ROS are the critical mediators underlying glucose-mediated epigenetic changes inducing NF- κ B activation, a process that leads to the expression of proinflammatory molecule VCAM-1 and MCP-1, and the activation of p66Shc by PKC β II [46,47]. Moreover, the protection of blood vessels from hyperglycemia-induced endothelial dysfunction exerted by SIRT1 occurs through a mechanism involving the negative regulation of p66Shc

[11,44,48,49]. The persistent p66Shc upregulation and mitochondrial translocation, which continues after returning to normoglycemia, is associated with continued ROS production, apoptosis, and reduced NO bioavailability [47]. The evidence, here provided, that the antioxidant and cytoprotective effects of Egt involves a positive regulation of SIRT1 and SIRT6 expression, add a new dimension to the mechanism underlying the role of Egt in protecting EC against hyperglycaemia-induced oxidative stress and senescence. At the same time, the intriguing novelty of our findings is strengthened by the observations that the effects of Egt includes the modulation of the cross-talk signaling between SIRT1/6 and p66Shc and NF- κ B, suggesting that this compound acts through complex dynamics of SIRT1- and SIRT6-regulated patterns. The SIRT1 and SIRT6-dependent protective effect of Egt further supports the established role of these sirtuins in the protection of endothelial cells from premature senescence and hGluc-induced senescence [50,51]. Within this scenario, it could be of interest to determine whether SIRT1 and SIRT6 act synergistically or in a distinct way to maintain endothelial homeostasis.

5. Conclusions

Given the fundamental role of oxidative stress in the onset of diabetes and its vascular complications, there is increasing interest in dietary supplements, natural extracts, polyphenols, and antioxidants to prevent vascular diseases by specifically modulating sirtuins [11]. Egt, obtained only from diet, is avidly accumulated, up to millimolar concentrations, in many cell types of human organism which possess a specific carrier for this substance. Egt presents high water solubility, stability at physiological pH, and antioxidants and metal chelating properties. Most of all, the newly identified redox pathway of Egt highlights a peculiar antioxidant role which accounts for the efficacy of this compounds in counteracting cell oxidative stress when compared to alkylthiols and other intracellular antioxidants. Furthermore, the widespread occurrence in food makes Egt a rich and free source of reducing equivalents, utilized continuously under basal cellular conditions and, at the same time, ensures additional protection during intensive ROS accumulation. In addition, the absence of toxicity at mM range of intracellular concentration and its ability to interfere with processes linked to diabetes through multiple pathways involving SIRT1 and SIRT6 regulation confer to Egt a clear role as dietary bioactive agent in the prevention of CVD associated with oxidative stress.

References

- [1] M. Félétou, P.M. Vanhoutte, Endothelial dysfunction: a multifaceted disorder (The Wiggers Award Lecture), *Am. J. Physiol. Heart Circ. Physiol.* 291 (2006) H985–1002.
- [2] A. Liguori, F.P. D'Armiento, A. Palagiano, M.L. Balestrieri, S. Williams-Ignarro, F. de Nigris, L.O. Lerman, M. D'Amora, M. Rienzo, C. Fiorito, L.J. Ignarro, W. Palinski, C. Napoli, Effect of gestational hypercholesterolaemia on omental vasoreactivity, placental enzyme activity and transplacental passage of normal and oxidised fatty acids, *BJOG* 114 (2007) 1547–1556.
- [3] Y. Mikhed, A. Daiber, S. Steven, Mitochondrial oxidative stress, mitochondrial DNA damage and their role in age-related vascular dysfunction, *Int. J. Mol. Sci.* 16 (2015) 15918–15953.
- [4] M.L. Balestrieri, M. Rienzo, F. Felice, R. Rossiello, V. Grimaldi, L. Milone, A. Casamassimi, L. Servillo, B. Farzati, A. Giovane, C. Napoli, High glucose downregulates endothelial progenitor cell number via SIRT1, *Biochim. Biophys. Acta* 2008 (1784) 936–945.
- [5] E.C. Eringa, E.H. Serne, R.I. Meijer, C.G. Schalkwijk, A.J. Houben, C.D. Stehouwer, Y.M. Smulders, V.W. van Hinsbergh, Endothelial dysfunction in (pre) diabetes: characteristics, causative mechanisms, and pathogenic role in type 2 diabetes, *Rev. Endocr. Metab. Disord.* 14 (2013) 39–48.
- [6] A. Liguori, C. Fiorito, M.L. Balestrieri, E. Crimi, B. Bruzzese, S. Williams-Ignarro, M. D'Amora, L. Sommese, V. Grimaldi, P.B. Minucci, A. Giovane, B. Farzati, L. J. Ignarro, C. Napoli, Functional impairment of hematopoietic progenitor cells in patients with coronary heart disease, *Eur. J. Haematol.* 80 (2008) 258–264.
- [7] M. Albiero, N. Poncina, M. Tjwa, S. Ciciliot, L. Menegazzo, G. Ceolotto, S. Vigili de Kreutzenberg, R. Moura, M. Giorgio, P. Pellicci, A. Avogaro, G.P. Fadini, Diabetes causes bone marrow autonomic neuropathy and impairs stem cell mobilization via dysregulated p66Shc and Sirt1, *Diabetes* 63 (2014) 1353–1365.
- [8] M.L. Balestrieri, L. Servillo, A. Esposito, N. D'Onofrio, A. Giovane, R. Casale, M. Barbieri, P. Paolisso, M.R. Rizzo, G. Paolisso, R. Marfella, Poor glycaemic control in type 2 diabetes patients reduces endothelial progenitor cell number by influencing SIRT1 signaling via platelet-activating factor receptor activation, *Diabetologia* 56 (2013) 162–172.
- [9] M.L. Balestrieri, M.R. Rizzo, M. Barbieri, P. Paolisso, N. D'Onofrio, A. Giovane, M. Siniscalchi, F. Minicucci, C. Sardù, D. D'Andrea, C. Mauro, F. Ferraraccio, L. Servillo, F. Chirico, P. Caiazzo, G. Paolisso, R. Marfella, Sirtuin 6 expression and inflammatory activity in diabetic atherosclerotic plaques: effects of incretin treatment, *Diabetes* 64 (2015) 1395–1406.
- [10] S. Zhou, H.Z. Chen, Y.Z. Wan, Q.J. Zhang, Y.S. Wei, S. Huang, J.J. Liu, Y.B. Lu, Z. Q. Zhang, R.F. Yang, R. Zhang, H. Cai, D.P. Liu, C.C. Liang, Repression of P66Shc expression by SIRT1 contributes to the prevention of hyperglycemia-induced endothelial dysfunction, *Circ. Res.* 109 (2011) 639–648.
- [11] N. D'Onofrio, M. Vitiello, R. Casale, L. Servillo, A. Giovane, M.L. Balestrieri, Sirtuins in vascular diseases: emerging roles and therapeutic potential, *Biochim. Biophys. Acta* 2015 (1852) 1311–1322.
- [12] M.L. Balestrieri, C. Fiorito, E. Crimi, F. Felice, C. Schiano, L. Milone, A. Casamassimi, A. Giovane, V. Grimaldi, V. del Giudice, P.B. Minucci, F. P. Mancini, L. Servillo, F.P. D'Armiento, B. Farzati, C. Napoli, Effect of red wine antioxidants and minor polyphenolic constituents on endothelial progenitor cells after physical training in mice, *Int. J. Cardiol.* 126 (2008) 295–297.
- [13] Q. Yuan, L. Chen, D.X. Xiang, Y.J. Li, C.P. Hu, Effect of resveratrol derivative BTM-0512 on high glucose-induced dysfunction of endothelial cells: role of SIRT1, *Can. J. Physiol. Pharmacol.* 89 (2011) 713–722.
- [14] L. Servillo, N. D'Onofrio, L. Longobardi, I. Sirangelo, A. Giovane, D. Cautela, D. Castaldo, A. Giordano, M.L. Balestrieri, Stachydrine ameliorates high-glucose induced endothelial cell senescence and SIRT1 downregulation, *J. Cell. Biochem.* 114 (2013) 2522–2530.
- [15] L.B. Gano, A.J. Donato, H.M. Pasha, C.M. Hearon Jr, A.L. Sindler, D.R. Seals, The SIRT1 activator SRT1720 reverses vascular endothelial dysfunction, excessive superoxide production and inflammation with aging in mice, *Am. J. Physiol. Heart Circ. Physiol.* 307 (2014) H1754–H1763.
- [16] C. Tanret, Sur une base nouvelle retiree du seigle ergote, l'ergothioneine, *Compt. Rend.* 149 (1909) 222–224.
- [17] R.C. Fahey, Novel thiols of prokaryotes, *Annu. Rev. Microbiol.* 55 (2001) 333–356.
- [18] C. Pfeiffer, T. Bauer, B. Surek, E. Schomig, D. Grundemann, Cyanobacteria produce high levels of ergothioneine, *Food Chem.* 129 (2011) 1766–1769.
- [19] I.K. Cheah, B. Halliwell, Ergothioneine: antioxidant potential, physiological function and role in disease, *Biochim. Biophys. Acta* 2012 (1822) 784–793.
- [20] H. Song, W. Hu, N. Naowarajna, A.S. Her, S. Wang, R. Desai, L. Qin, X. Chen, P. Liu, Mechanistic studies of a novel C-S lyase in ergothioneine biosynthesis: the involvement of a sulfenic acid intermediate, *Sci. Rep.* 5 (2015) 11870.
- [21] L. Servillo, D. Castaldo, R. Casale, N. D'Onofrio, A. Giovane, D. Cautela, M. L. Balestrieri, An uncommon redox behavior sheds light on the cellular antioxidant properties of ergothioneine, *Free Radic. Biol. Med.* 79 (2015) 228–236.
- [22] R.W. Li, C. Yang, A.S. Sit, Y.W. Kwan, S.M. Lee, M.P. Hoi, S.W. Chan, M. Hausman, P.M. Vanhoutte, G.P. Leung, Uptake and protective effects of ergothioneine in human endothelial cells, *J. Pharmacol. Exp. Ther.* 350 (2014) 691–700.
- [23] I. Kaneko, Y. Takeuchi, Y. Yamaoka, Y. Tanaka, T. Fukuda, Y. Fukumori, T. Mayumi, T. Hama, Quantitative determination of ergothioneine in plasma and tissues by TLC densitometry, *Chem. Pharm. Bull.* 28 (1980) 3093–3097.
- [24] K.R. Martin, The bioactive agent ergothioneine, a key component of dietary mushrooms, inhibits monocyte binding to endothelial cells characteristic of early cardiovascular disease, *J. Med. Food* 13 (2010) 1340–1346.
- [25] A. Arduini, L. Eddy, P. Hochstein, The reduction of ferryl myoglobin by ergothioneine: a novel function for ergothioneine, *Arch. Biochem. Biophys.* 281 (1990) 41–43.
- [26] O. Sakrak, M. Kerem, A. Bedirli, H. Pasaoglu, N. Akyurek, E. Ofluoglu, F. A. Gultekin, Ergothioneine modulates proinflammatory cytokines and heat shock protein 70 in mesenteric ischemia and reperfusion injury, *J. Surg. Res.* 144 (2008) 36–42.
- [27] N. Arkadopoulos, C. Nastos, K. Kalimeris, E. Economou, K. Theodoraki, E. Kouskouni, A. Pafiti, G. Kostopanagiotou, V. Smyrniotis, Iron chelation for amelioration of liver ischemia–reperfusion injury, *Hemoglobin* 34 (2010) 265–277.
- [28] R.M. Eppard, The role of dietary ergothioneine in the development of diabetes mellitus, *Med. Hypotheses* 9 (1982) 207–213.
- [29] R. Fraser, S. Jegard, Blood ergothioneine levels in diabetes mellitus, *J. Lab. Clin. Med.* 35 (1950) 960–967.
- [30] C.H. Beatty, Levels of ergothioneine in the red blood cell in experimental diabetes, *J. Biol. Chem.* 199 (1952) 553–561.
- [31] M.V. Guizarro, A. Indart, O.I. Aruoma, M. Viana, B. Bonet, Effects of ergothioneine on diabetic embryopathy in pregnant rats, *Food Chem. Toxicol.* 40 (2002) 1751–1755.
- [32] Q. Yuan, J. Peng, S.Y. Liu, C.J. Wang, D.X. Xiang, X.M. Xiong, C.P. Hu, Y.J. Li, Inhibitory effect of resveratrol derivative BTM-0512 on high glucose-induced cell senescence involves dimethylaminohydrolase/asymmetric dimethylarginine pathway, *Clin. Exp. Pharmacol. Physiol.* 37 (2010) 630–635.
- [33] H.Z. Chen, Y.Z. Wan, D.P. Liu, Cross-talk between SIRT1 and p66Shc in vascular diseases, *Trends Cardiovasc. Med.* 23 (2013) 237–241.
- [34] T.L. Kawahara, E. Michishita, A.S. Adler, M. Damian, E. Berber, M. Lin, R. A. McCord, K.C. Ongaigui, L.D. Boxer, H.Y. Chang, K.F. Chua, SIRT6 links histone H3 lysine 9 deacetylation to NF-kappaB-dependent gene expression and organismal life span, *Cell* 136 (2009) 62–74.
- [35] M. Lappas, Anti-inflammatory properties of sirtuin 6 in human umbilical vein endothelial cells, *Mediat. Inflamm.* 2012 (2012) 597514.
- [36] R. Liu, H. Liu, Y. Ha, R.G. Tilton, W. Zhang, Oxidative stress induces endothelial cell senescence via downregulation of Sirt6, *Biomed. Res. Int.* 2014 (2014) 902842.
- [37] Y.M. Go, D.P. Jones, Cysteine/cystine redox signaling in cardiovascular disease, *Free Radic. Biol. Med.* 50 (2011) 495–509.
- [38] A. Arduini, G. Mancinelli, G.L. Radatti, P. Hochstein, E. Cadenas, Possible mechanism of inhibition of nitrite-induced oxidation of oxyhemoglobin by L-Ergothioneine and uric acid, *Arch. Biochem. Biophys.* 294 (1992) 398–402.
- [39] F. Misiti, M. Castagnola, C. Zuppi, B. Giardina, I. Messina, Role of ergothioneine on S-nitrosoglutathione catabolism, *Biochem. J.* 356 (2001) 799–804.
- [40] A.S. Arora, G.J. Gores, The role of metals in ischemia/reperfusion injury of the liver, *Semin. Liver Dis.* 16 (1996) 31–38.
- [41] K.T. Spencer, P.D. Lindower, G.R. Buettner, R.E. Kerber, Transition metal chelators reduce directly measured myocardial free radical production during reperfusion, *J. Cardiovasc. Pharmacol.* 32 (1998) 343–348.
- [42] S. Sotgia, E. Pisanu, G. Pintus, G.L. Erre, G.A. Pinna, L. Deiana, C. Carru, A. Zinellu, Plasma L-ergothioneine measurement by high-performance liquid chromatography and capillary electrophoresis after a pre-column derivatization with 5-iodoacetamidofluorescein (5-IAF) and fluorescence detection, *PLoS One* 8 (2013) e70374.
- [43] R.H. Houtkooper, E. Pirinen, J. Auwerx, Sirtuins as regulators of metabolism and healthspan, *Nat. Rev. Mol. Cell Biol.* 13 (2012) 225–238.
- [44] L.F. Chen, W.C. Greene, Shaping the nuclear action of NF-kB, *Nat. Rev. Mol. Cell Biol.* 5 (2004) 392–401.
- [45] F. Yeung, J.E. Hoberg, C.S. Ramsey, M.D. Keller, D.R. Jones, R.A. Frye, M.W. Mayo, Modulation of NF-kB-dependent transcription and cell survival by the SIRT1 deacetylase, *EMBO J.* 23 (2004) 2369–2380.
- [46] A. Magenta, S. Greco, M.C. Capogrossi, C. Gaetano, F. Martelli, Nitric oxide, oxidative stress, and p66Shc interplay in diabetic endothelial dysfunction, *Biomed. Res. Int.* 2014 (2014) 193095.
- [47] F. Paneni, P. Mocharla, A. Akhmedov, S. Costantino, E. Osto, M. Volpe, T. F. Lüscher, F. Cosentino, Gene silencing of the mitochondrial adaptor p66(Shc) suppresses vascular hyperglycemic memory in diabetes, *Circ. Res.* 111 (2012) 278–289.
- [48] E. Migliaccio, M. Giorgio, S. Mele, G. Pelicci, P. Reboldi, P.P. Pandolfi, L. Lanfranccone, P.G. Pelicci, The p66Shc adaptor protein controls oxidative stress response and life span in mammals, *Nature* 402 (1999) 309–313.
- [49] G.G. Camici, M. Schiavoni, P. Francia, M. Bachschmid, I. Martin-Padura, M. Hersberger, F.C. Tanner, P. Pelicci, M. Volpe, P. Anversa, T.F. Lüscher, F. Cosentino, Genetic deletion of p66(SHC) adaptor protein prevents hyperglycemia-induced endothelial dysfunction and oxidative stress, *Proc. Natl. Acad. Sci. USA* 104 (2007) 5217–5222.
- [50] G. Arunachalam, S.M. Samuel, I. Marei, H. Ding, C.R. Triggle, Metformin modulates hyperglycemia-induced endothelial senescence and apoptosis through SIRT1, *Br. J. Pharmacol.* 171 (2014) 523–535.
- [51] A. Cardus, A.K. Uryga, G. Walters, J.D. Erusalimsky, SIRT6 protects human endothelial cells from DNA damage, telomere dysfunction, and senescence, *Cardiovasc. Res.* 97 (2013) 571–579.